Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 – 2024

Source: Global Carbon Budget 2025 · NASA/NOAA

CO_2

429 ppm

Feb 2026 Mauna Loa Reading

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Atmospheric CO_2 (Feb 2026)

280 ppm

Pre-Industrial Baseline (˜1750)

+53%

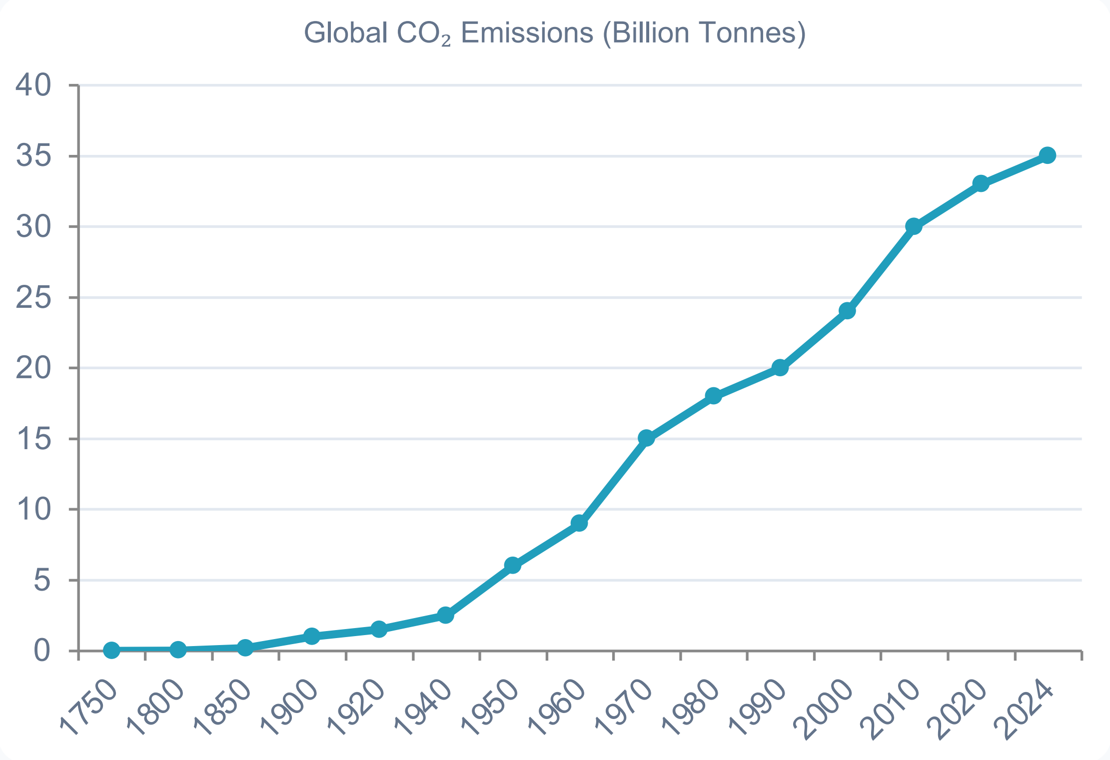
Increase since pre-industrial era

Key Context

- Current level is 50%+ above the ˜280 ppm pre-industrial baseline
- Rate of increase is at least 1.5× faster than any natural ice-age transition
- Sharp acceleration since industrialization — primarily from coal, oil, and gas combustion
- Ground-based (Mauna Loa) and satellite measurements confirm the trend

Source: NOAA Mauna Loa Observatory · Global Carbon Budget 2025

Global Fossil Fuel CO_2 Grew Six-Fold: 6 Billion to 35+ Billion Tonnes (1950–2024)



Key Inflection Points

1950: ˜6 Bt
Pre-WWII baseline

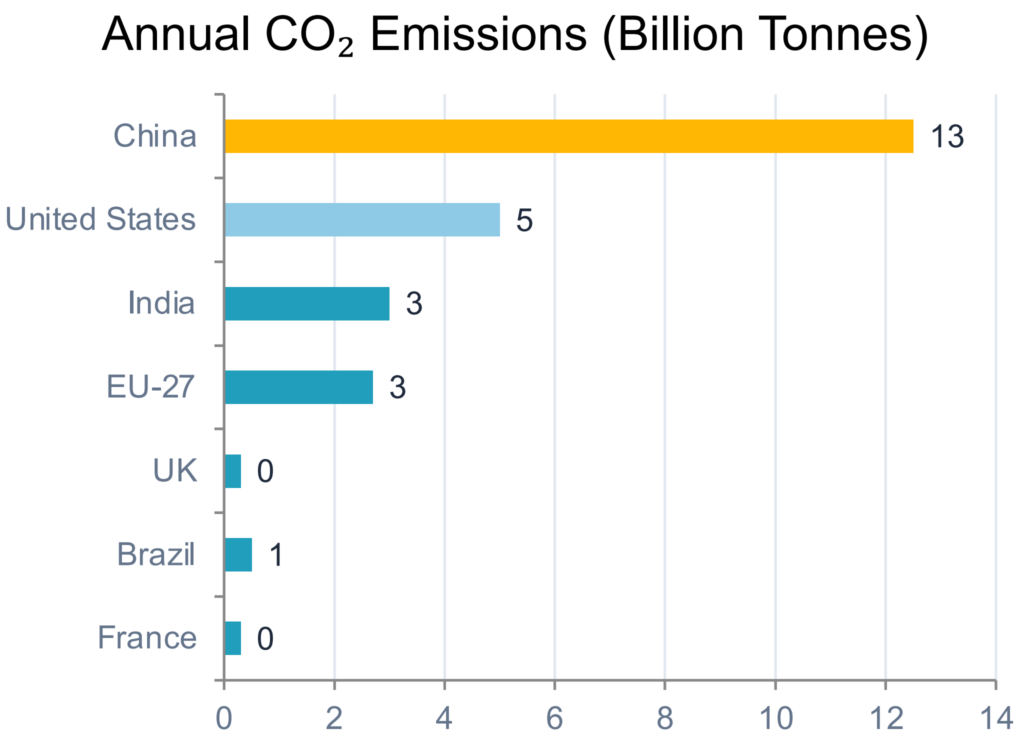
1990: ˜20 Bt
Nearly 4× the 1950 level

2024: 35+ Bt
Six-fold increase since 1950

Growth has slowed but not peaked

Source: Global Carbon Budget 2025

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



Key Observations

- China: ~12.5 Bt — over 27% of global total, slight upward trend
- USA: ~5 Bt — peaked near 6 Bt around 2005, now declining
- India: ~3 Bt — rising rapidly, third-largest emitter
- EU-27: ~2.7 Bt — declining, less than 7% of global total
- UK and France each ~0.3 Bt — under 1% each

See also: Figure 3 — Annual CO_2 by country

Source: Global Carbon Budget 2025

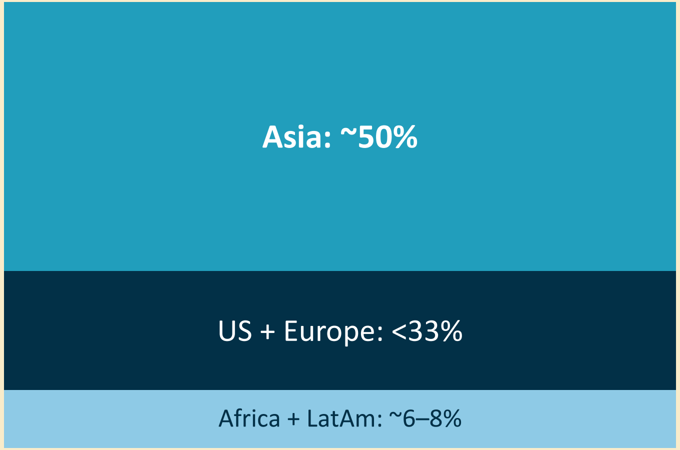
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

Emission Share by Region

2024

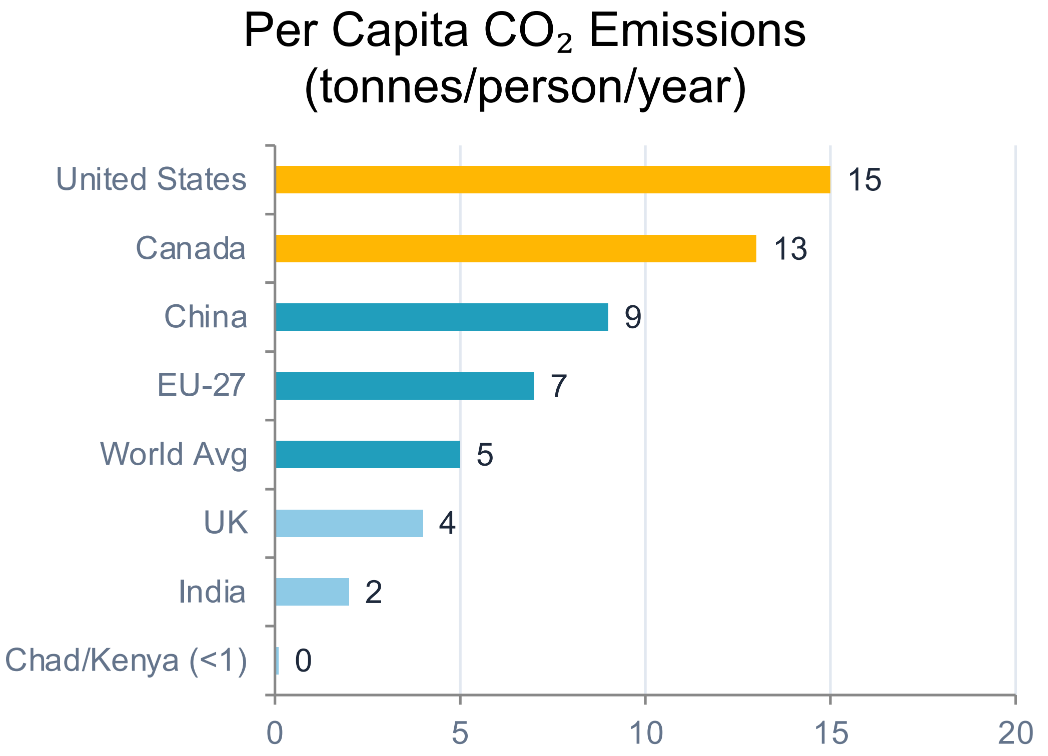
Emission Share by Region



Africa and South America each contribute only 3–4% — comparable to international aviation & shipping

Source: Global Carbon Budget 2025 · Our World in Data

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



150×

Gap between US/Australia and Chad/Niger

Key Insight

- US & Australia: ~15 t/person — ~3× the global average
- China: ~9 t/person — near global average, rapidly rising
- India: ~2 t/person — well below global average of ~5 t
- Chad/Niger: ~0.1 t — average American emits this in under 2 days

See also: Figure 1 — Per capita CO_2 over time

Source: Global Carbon Budget 2025 · Our World in Data

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

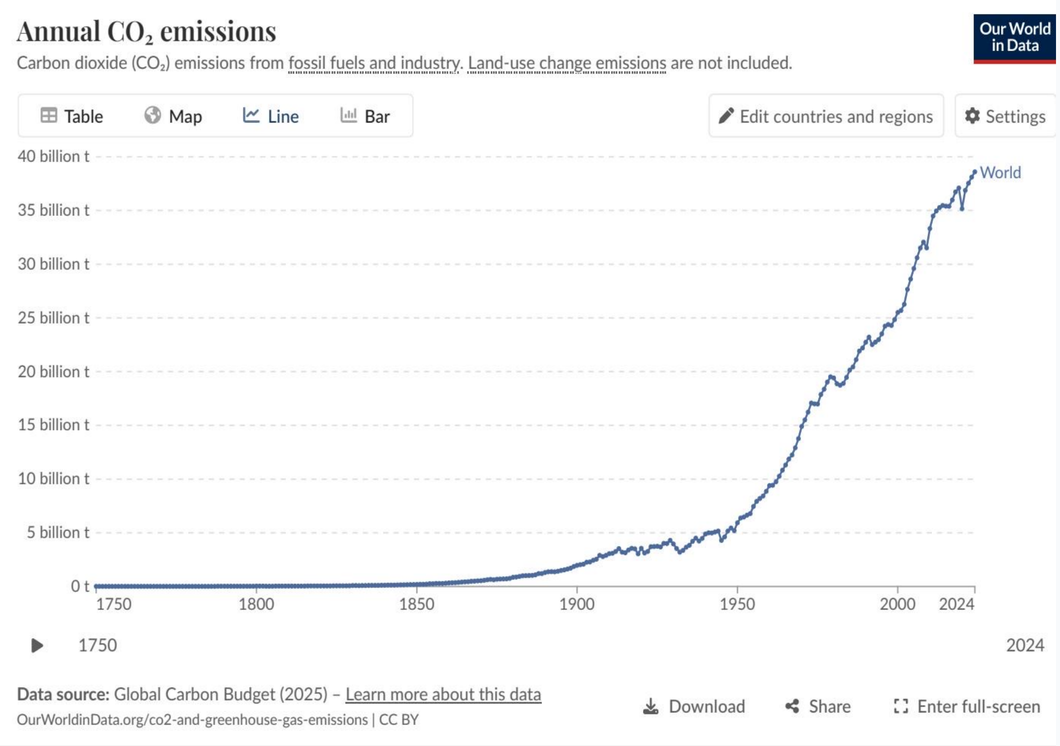


Figure 4: Annual % change in CO_2 emissions, 2024

Regional Trends

Declining (Blue)

- US: −1% to −5%
- Most of Europe: −1% to −5%

Increasing (Orange/Red)

- Russia: +2% to +10%+
- Parts of SE Asia & Africa: +2% to +10%
- Some S. America: >+10% (isolated)

Mixed picture globally — no uniform trajectory

Source: Global Carbon Budget 2025 · Figure 4

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

France & UK

Per Capita CO_2: ~4 t/person

- High nuclear power share (France: ~70% nuclear)
- Growing renewables contribution
- Lower reliance on fossil fuels for electricity
- Result: among lowest per-capita in developed world

Germany & Netherlands

Per Capita CO_2: ~7–9 t/person

- Higher reliance on coal and natural gas
- Renewables growing but transition still underway
- Germany phasing out nuclear — increased gas use
- Netherlands: gas-heavy energy system

Country	Per Capita CO_2 (t/yr)	Key Energy Source	Trend
France	~4	Nuclear (~70%) + Renewables	Declining
United Kingdom	~4	Gas + Growing Renewables	Declining
Germany	~7–9	Coal + Gas (Nuclear phase-out)	Declining slowly
Netherlands	~8	Natural Gas dominant	Declining slowly

Source: Global Carbon Budget 2025 · Our World in Data

Key Takeaways

Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

1. Global emissions exceed 35 Bt/yr and have not yet peaked, despite slowing growth
2. China alone accounts for >25% of global total (~12.5 Bt) — more than double the US (~5 Bt); India (~3 Bt) is third and rising
3. Geographic center of emissions has dramatically shifted — from >90% in Europe+US in 1900 to Asia producing ~50% today
4. Per capita inequalities are extreme — a 150× gap separates US/Australia (~15 t) from Chad/Niger (~0.1 t)
5. Energy mix and policy choices — not just prosperity — determine per capita outcomes; France vs. Germany shows this clearly